LESSON 2.5

Describe Data by Range and Interquartile Range

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

WHY THIS MATTERS The league analyst is comparing two sixth-grade classes that took the same math test. Both classes have the SAME median score, 81 — so the medians alone cannot tell them apart. Your job is to measure how spread out each class's scores are, using the range and the interquartile range, and decide which class did better overall.

REMEMBER

Describe Data by Range and Interquartile Range

Measures of Variability: Range & IQR. Formula: Range = Maximum − Minimum | IQR = Q3 − Q1 A measure of center like the median tells you where the data sits. It does not tell you how far apart the values are. Two classes can share the same median and still look nothing alike. Range and interquartile range are the two numbers that describe that spread.

1. Range measures the total spread across the entire data set.
2. IQR (Interquartile Range) measures the spread of the middle 50% of data.
3. Both are non-negative distance measures of variability

Words: **Range and Interquartile Range *(Rango y rango intercuartílico)*** **Range *(Rango)*** **Quartile *(Cuartil)*** **Interquartile range *(Rango intercuartílico)*** **Variability *(Variabilidad)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Class B's nine retest scores are 72, 76, 78, 80, 80, 80, 82, 84, 88. Which statement describes the center, spread, AND shape of this distribution?

(A) Center about 80, range 16, roughly symmetric

(B) Center about 80, range 88, roughly symmetric

(C) Center about 16, range 80, roughly symmetric

(D) Center about 80, range 16, skewed right because 88 is the greatest score

HINT Center: the middle (5th) score once they are in order.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Both classes have a median score of 81. Class A's scores make a symmetric dot plot. Class B's scores pile up between 78 and 86, with one score of 52. Which statement about Class B is true?

(A) Class B's distribution must have the same shape as Class A's because their medians are equal

(B) Class B's distribution is skewed right because 52 is far away from the other scores

(C) Class B's distribution is skewed left: the single low score of 52 makes a tail toward the lower scores

(D) Class B's distribution is symmetric because most of its scores are close to 81

HINT Find where most of Class B's scores pile up.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

In that same box plot, what does the LEFT edge of the box show?

(A) Q1, which is 7

(B) Q3, which is 14

(C) The minimum, which is 4

(D) The median, which is 10

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A box plot has minimum 4, Q1 = 7, median = 10, Q3 = 14 and maximum 20. Which value splits the data in half?

(A) 7

(B) 10

(C) 12

(D) 14

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 **Circle the letter of the best answer. Show how you know.**

Find the five-number summary of 2, 5, 6, 8, 9, 11, 14.

(A) 2, 5, 8, 11, 14
(B) 2, 6, 8, 9, 14
(C) 2, 5, 6, 11, 14
(D) 2, 8, 11, 14, 14

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

About what fraction of the data lies inside the box of a box plot?

(A) About one quarter
(B) All of it
(C) About half
(D) None of it

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

SENTENCE STARTER I would use the ___ because it only uses the ___, so one unusual value cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each phrase under the correct measure.

Groups: Range | Interquartile range

_____ Uses all the data

_____ Uses the middle half

_____ Greatest − least

_____ Q3 − Q1

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Two sixth-grade classes took the same math test. Class A's box plot reads least 62, Q1 74, median 81, Q3 87, greatest 90. Class B's reads least 70, Q1 76, median 81, Q3 84, greatest 90. Both classes have a median of 81, so the median alone cannot tell you which class did better overall. What is the range of Class A's scores?

(A) 13 points

(B) 28 points

(C) 81 points

(D) 90 points

Put the data in order _____

WORK

3 **Circle the letter of the best answer. Show how you know.**

Two sixth-grade classes took the same math test. Class A's box plot reads least 62, Q1 74, median 81, Q3 87, greatest 90. Class B's reads least 70, Q1 76, median 81, Q3 84, greatest 90. Both classes have a median of 81, so the median alone cannot tell you which class did better overall. What is the interquartile range of Class B's scores?

(A) 8 points
(B) 14 points
(C) 20 points
(D) 81 points

Put the data in order _____

WORK

4 **Circle the letter of the best answer. Show how you know.**

Two sixth-grade classes took the same math test. Class A's box plot reads least 62, Q1 74, median 81, Q3 87, greatest 90. Class B's reads least 70, Q1 76, median 81, Q3 84, greatest 90. Both classes have a median of 81, so the median alone cannot tell you which class did better overall. Both classes have a median of 81. What do the smaller range and smaller IQR tell you about Class B?

(A) Class B's best score was higher
(B) Class B had fewer students
(C) Class B's median is actually larger
(D) Class B's scores are less spread out, so more students scored near the median

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the range of 7, 9, 12, 15, 20?

(A) 7
(B) 9
(C) 13
(D) 20

Put the data in order _____

WORK

6 **Circle the letter of the best answer. Show how you know.**

A box plot has Q1 = 30 and Q3 = 45. What is the IQR?

(A) 15

(B) 37.5

(C) 45

(D) 75

Put the data in order _____

WORK

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

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